Response to the CMS/ASTP RFI Health Technology Ecosystem
CMS-0042-NC

June 16, 2025

Centers for Medicare & Medicaid Services
Department of Health and Human Services
Attention: CMS-2025-0050-0031
P.O. Box 8013
Baltimore, MD 21244–8013

Regarding CMS-2025-0050-0031, Request for Information (RFI) Health Technology Ecosystem

To the Director of the Centers for Medicare & Medicaid Services (CMS):

The Association of Public Health Laboratories (APHL) is pleased to provide this response to the CMS/ASTP Request for Information: Health Technology Ecosystem. We concentrate on two main points in our response: 1) TEFCA is a necessary framework that is working in today's digital health ecosystem and 2) our innovative technical platform (AIMS) is a successful example of the scalability of TEFCA.

Over the last decade, TEFCA has fundamentally changed the expectations for how healthcare and its partners store, manage, and share data. While TEFCA is still a work in progress, this common agreement has allowed hospitals, payers, public health, and health IT vendors to build towards a common goal and a recognized set of functional requirements and data exchange standards.

APHL has observed the challenges and burden of managing data exchange when partners do not conform to the same standards or data reporting requirements. These barriers ultimately impact provider/patient access to health information and healthcare quality metrics like time-to-diagnosis and time-to-treat and the ability for public health to respond to emerging infectious disease outbreaks. TEFCA combines the lessons learned through past national interoperability initiatives to provide clear blueprints to health data exchange partners. Expansion of TEFCA should continue.

It will take time and investment, as well as commitment and leadership from federal partners, especially ASTP, for TEFCA to reach full maturity. While pilot engagements between QHIN participants are encouraging, the number of participating hospitals, payers, and other entities in production on TEFCA must reach a critical mass for the true potential of this framework to be realized. Further, these partners need a comprehensive library of Exchange Purposes and sub-exchange purposes so they can implement interoperable data exchange across many use cases. They also need to be incentivized to continue to develop

and pilot robust technology solutions that streamline and simplify data exchange – and ensure that this exchange delivers high-quality, timely, usable data.

The APHL Informatics Messaging Services (AIMS) Platform is an instructive example of an early TEFCA success founded on innovative technology, domain expertise, and public/private partnerships. While hospitals are still onboarding to QHINs, AIMS has already established connections to 70+ state, local, and territorial public health agencies (PHAs) and currently supports 55 of these PHAs to receive electronic case reporting (eCR) via the eHealth Exchange QHIN, one of the first examples of public health data in production through TEFCA. AIMS is processing 2.25 million eCR documents via TEFCA each month. In July 2024, AIMS began sending anonymized eICR/RR FHIR bundles to CDC (for PHAs that approve).

Beyond TEFCA, the AIMS Platform securely transports millions of messages daily for use cases that span the full spectrum of public health data exchange. This data exchange is critical both for patient care and for national security and emergency preparedness. Laboratory test results are needed to conduct differential diagnoses, identify specific conditions, and start and manage treatment, while public health reporting ensures that state, local, and federal programs have the data needed to safeguard and improve the public's health, whether it be from infectious disease, foodborne disease outbreaks, chronic disease, cancer, bioterrorism or other public health concerns. AIMS routes data for these and other use cases, including eCR, electronic laboratory report (ELR), syndromic surveillance, electronic laboratory surveillance messages (ELSM), case notification messages, cancer reporting, vital records, immunization data, and electronic test order and result (ETOR). Crucially, through our collaboration with eHealth Exchange, as TEFCA finalizes sub-exchange purposes for these use cases, AIMS is positioned to seamlessly transition these data feeds to TEFCA for its data exchange partners.

"If we were to lose AIMS, that would be a DEFCON 1 disaster. There is no realistic alternative."

Reduced point-to-point connections

The AIMS hub model has reduced the need for redundant system integrations and multiple point-to-point connections by serving as the intermediary between data exchange partners.

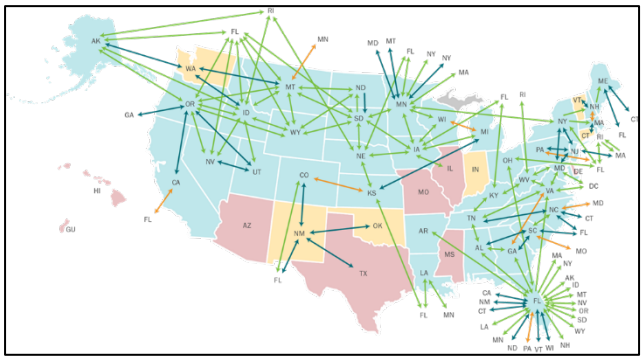
ELR with clinical laboratories

Quest manages at least 3 laboratory information management systems (LIMS) across its facilities. If each LIMS had to connect to each PHA, complete coverage would potentially require the onboarding, testing and continued maintenance of more than 150 point-to-point connections, each with its own messaging requirements and format. Quest needed a streamlined solution to onboard its laboratories to send ELR to PHAs that could be maintained with limited overhead. Quest Diagnostics partnered with APHL to ELR through the AIMS Platform to all public health agencies (PHAs). Now, Quest sends a single feed to AIMS via VPN, where each message is validated and routed to the appropriate PHA within the connected network using each PHA's preferred transport protocol.

This public/private partnership yielded significant cost-savings for Quest and the cessation of onerous manual reporting. The project simultaneously improved the quality and timeliness of the data reported, enabling public health to quickly take action on potential health threats.

Cross-jurisdictional data exchange

Once onboarded to AIMS, our partners can easily add more use cases and dataflows with other partners. As individuals move around the country, to travel or to be seen by a provider, public health often needs to share data about residents of other states. Dozens of public health agencies use our InterPartner service to exchange data with other agencies, while still maintaining a single connection to AIMS.



InterPartner Connections on AIMS

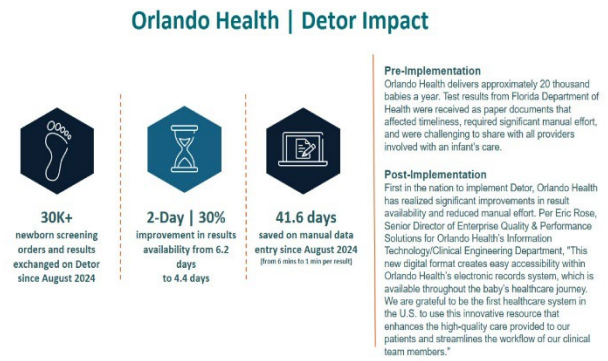
Agility in a public health emergency

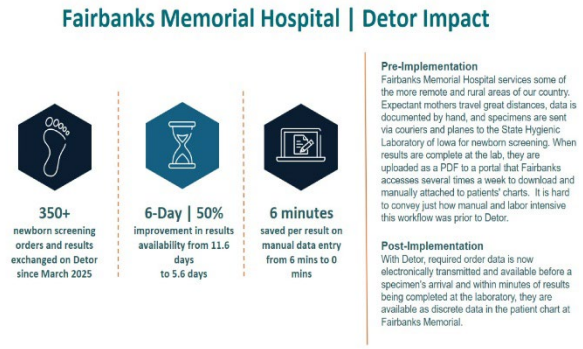
Partners exchanging data via the AIMS Platform are able to add conditions and use cases quickly and efficiently, even during a public health response. For example, Quest used its existing connection with AIMS to become one of the first private laboratories to send COVID data to State PHAs, CDC, and Health and Human Services (HHS). Adding COVID, and later mpox, to the Quest ELR feed required relatively little effort on the part of either the laboratory or the PHAs.

Economy of Scale

AIMS is not just a hub; it performs validation, transformation and translation services. Centralizing these services reduces the costs for ongoing operations and maintenance for all partners. Data is standardized by AIMS and thus consumable by the systems and applications that must use that data. In some cases, the burden to "clean" data is nearly eliminated, further ensuring that providers and patients can receive their laboratory test results as soon as they are released. When providers have access to this information immediately, the time to diagnosis and time to treat is improved.

Many of the services and solutions that AIMS provides address critical needs in public health data exchange that agencies have struggled to implement locally. AIMS provides the centralized solution, the technology, and the technical assistance to implement and maintain it, allowing public health agencies to achieve milestones that have long been on their roadmap. For example, public health laboratories have long recognized the efficiencies of ETOR, including cost and time-savings, reduced data entry, higher quality data, faster turnaround times, and improved forecasting, but have had limited success implementing ETOR solutions. It is technologically challenging and cost-prohibitive to build their own solution and then onboard each hospital one by one with limited resources. Offering a trusted intermediary, Detor, on AIMS has accelerated the adoption of ETOR with public health





starting with newborn screening results, leading to better outcomes for the youngest patients in the healthcare system. Detor's validation, translation, and transformation services are centrally hosted, and we work with the hospital and the laboratory during the onboarding process, making the implementation faster and easier for all parties. And once onboarded, hospitals can share ETOR with other laboratories that are part of the Detor program. We now have a queue of hospitals and laboratories eager to onboard to Detor.

Public health domain expertise

"AIMS is not just technology but also a unique arrangement of data partners."

APHL understands public health practice, standards, and requirements. Our subject matter experts have the specialized knowledge to implement solutions that meet the particular demands of public health data exchange. Moreover, we have established a relationship of trust with the entire public health community at the local, state, territorial, and federal level, and increasingly with private industry partners as well. By serving as a public health intermediary, the AIMS Platform eliminates the need for one-off solutions for public health data exchange, reduces redundant systems integrations and centralizes standardization and validation services. Through our partnership with eHealth Exchange, the AIMS Platform will be able to connect to all QHINs and will be able to provide these services to any member of the TEFCA network. AIMS takes the burden of interoperable data exchange off public health and healthcare practitioners and lets them focus on what's important, while also ensuring that decision-makers have access to better, more complete higher-quality data. The AIMS Platform is a cornerstone of the nation's modernized health IT infrastructure, and with TEFCA, can bring crucial data quickly to where they are necessary.

Future Services

AIMS has plans to develop additional shared services that will benefit the entire digital health ecosystem, improve public health outcomes, and allow healthcare to focus on patient care rather than reporting requirements. The CDC DIBBS team is working with AIMS now to launch its eCR Refiner tool as a centralized service on the AIMS Platform to "refine" eCR messages and surface the relevant data before they are received by public health. In the future, AIMS could utilize the eCR model for ELR, routing lab results to the eligible PHA from all connected healthcare providers. AIMS has multiple pilot PHLs and HCOs in production with ETOR and is positioned to scale this service nationally. AIMS is also building its FHIR capabilities and has partnered with Deloitte and the Florida Department of Health to conduct a pilot project to run a FHIR query to retrieve missing patient

demographic data from an EHR. We have the infrastructure to scale these solutions simultaneously via TEFCA and separately from TEFCA, depending on the readiness of participants. In addition, AIMS possesses the functionality to perform robust validation on incoming data, and this technology could eventually be adapted to support CMS performance-based metrics around data quality and usability.

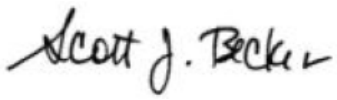
Conclusion

Thank you again for the opportunity to comment on CMS-0042-NC, Request for Information, Health Technology Ecosystem. We strongly encourage CMS and ASTP to continuing the visionary work initiated under TEFCA to build an integrated health IT ecosystem and to continue to engage public health leadership at all levels in this planning.

Furthermore, we encourage CMS and ASTP to consider that AIMS is a critical piece of the digital health ecosystem, a trusted intermediary with existing relationships with PHAs and healthcare and proven success as a TEFCA partner. Our subject matter experts understand public health standards and reporting requirements on a deep level, and we provide technical assistance and centralized services to senders and receivers. This model has consistently achieved economy of scale and an efficient use of resources. Let AIMS focus on what we do best: public health.

Should you have any questions about APHL's response, please contact Melanie Kourbage, Lead Specialist, APHL Informatics, at melanie.kourbage@aphl.org.

Sincerely,



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